

Speech Tech Week-7

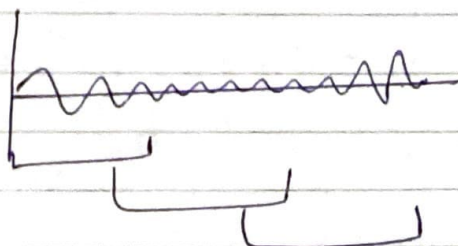
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L1 Continuous vs Discrete HMM Models

- Yes (3) & No (2 sounds)
 $z_1(t)$ $x_1(t)$ with labels

supervised learning

- complex than /ah/ & /iy/ problem.



O_1, O_2, O_3 (80x1) MFCC vectors
.....

- recall - HMM problem in weather conditions
- hidden states $\rightarrow$ 'y', 'ch', 's'
- observable states $\rightarrow O_1, O_2, \dots$ MFCC vectors
- model each of the state with the mixture of gaussians

GMM^y ($O_1; \Lambda_y$)

$$\text{params } \sum_{i=1}^{16} w_i N(x; \mu_i^y, \Sigma_i^y)$$

- having obs. $O_1, \dots, O_8$
what seq. of phonemes best explain this?

given the phoneme seq., the lexicon helps to map to the words.

L7.2 Different Types of HMM Models

- phonemes were the hidden states. They are not spoken in isolation.

cat → they
cat are
pat affect by both L & R sides.

- 3 state model for each phoneme

↳ each phoneme is modelled with 3 states -

1. transiⁿ from prev. sound
2. steady state
3. transiⁿ to next sound

each state has GMM modelling

- context dependency or triphone model
(context dependent phones) (triple phone)

→ The L & R side influence its behaviour.

Have a sep. based on the L & R context.

→ In spoken lang. every phoneme depending on L & R phoneme → we have a sep. model.

- Each phoneme has 3 states.

44 sounds/phonemes



3 state model

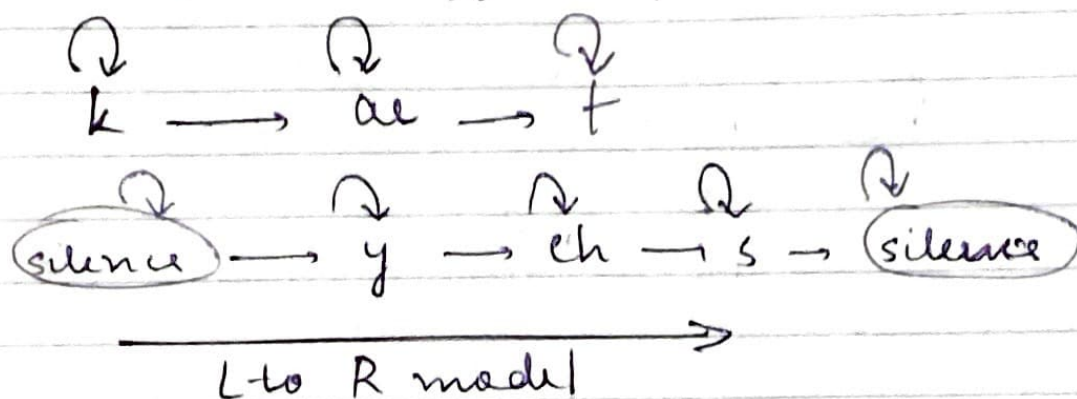
$$p \rightarrow [p_{(ah_1)}^{+n} \quad p_{(ah_2)}^{+m} \quad p_{(ah_3)}^{+n}] \rightarrow n$$

for phoneme, 43 → [ah] → 43

43x43 sep. models, each of them being 3 state model.

almost → 44³

- because of data-sufficiency $\rightarrow$ state-tying



- $ae \rightarrow$ monophone
 - $k + ae + t \rightarrow$ triphone
 - $k + ae \rightarrow$ diphone
 - $ae + t \rightarrow$ diphone
- models they could be context dependent or independent

L3 Lexicon / Pronunciation Dictionary

- language model $\rightarrow$ grammar
- lexicon $\rightarrow$ pronounced dictionary
 - $\rightarrow$ based on phones (40-60)
 - $\rightarrow$ written by human
 - $\rightarrow$ it is the link btw. sub-word HMMs & language models.
- $\rightarrow$ construction of a lexicon involves ensuring high coverage of words & phonetics expansion of each word.
- $\rightarrow$ it becomes a lookup table.
- $\rightarrow$ in most of the indian lang. $\rightarrow$ we write what we speak.

L7.4 Introduction to Language Model

- Task of predicting what word comes next

$$P(x^{(t+1)} | x^{(1)}, \dots, x^{(t)})$$

- assigning probability to piece of text (collection of words)

$$P(x^{(1)} \dots x^{(T)})$$

↪ joint distribution

$$P(x^1 \dots x^T) = \prod_{t=1}^T P(x^t | x^{t-1} \dots x^1)$$

relaⁿ with the help of bayes' rule

- LMs help a speech recognizer figure out how likely a word seq. is independent of the acoustics.
- other applicaⁿ → text gen, next word predicⁿ, QA problem, machine translaⁿ.
- ASR → finding most probable word seq. (w) given a speech sequence (s).

$$1. \tilde{W} = \operatorname{argmax}_w P(w/s)$$

↓ conditional probability th.

$$2. P(w/s) = \frac{P(s|w) P(w)}{P(s)}$$

$$3. \tilde{W} = \operatorname{argmax}_w \frac{P(s|w) P(w)}{P(s)}$$

this comes from acoustic model

→ this comes from a LM.

LS.5 N-Gram Lang. Model

- Ngram defined as a contiguous seq. of n tokens from a given piece of text. Some of them are unigram, bigrams, trigrams
(singles) (pairs) (triplets)
- $P(x^1 \dots x^T) = \prod_{t=1}^T P(x^t | x^{t-1} \dots x^1)$

x^t depends only in $n-1$ words

$$P(x^t | x^{t-1} \dots x^1) = P(x^t | \underbrace{x^{t-1} \dots x^{t-n+1}}_{(n-1) \text{ words}})$$

to calculate this, we use large corpus of text

prob. of n -gram $\leftarrow$ $= \frac{P(x^t, x^{t-1} \dots x^{t-n+1})}{P(x^{t-1} \dots x^{t-n+1})}$

prob. of $n-1$ gram $\leftarrow$

- But, N-gram lang. models are not able to capture long range dependencies.
- what if some sentence has never occurred in the data? $\Rightarrow$ add small δ to the count for every $w \in V$. This c/d smoothing.
- 2 or 3 words might not occur together $\Rightarrow$ just condiⁿ on (2-1) or (3-1) words instead. This c/d backoff.

- Perplexity

no. of words in the test set

$$PP(W) = N \sqrt[N]{\frac{1}{P(w_1, \dots, w_N)}}$$

perplexity for test set W

likelihood of test set W

- Evaluation of LM.

1. Extrinsic → using LM in actual task (ASR, etc.). Take the model which has SOTA

2. Intrinsic → using some kind of metrics.
→ perplexity

- if a model assign a high probab. to a test set, that means it's not surprised to see it.
High probab. → low perplexity and vice-versa.

L7.6 Intro. to Deep Neural Networks

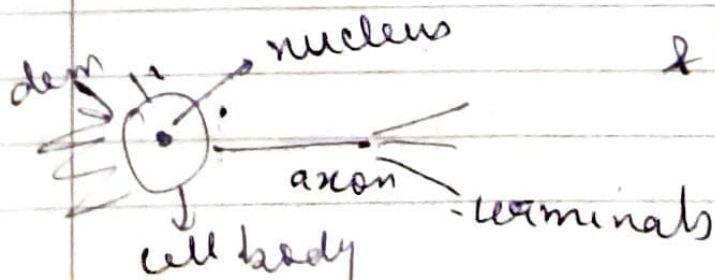
- neuron has input (dendrites) & output (axon)

- info. goes from dendrites to axon via cell body.

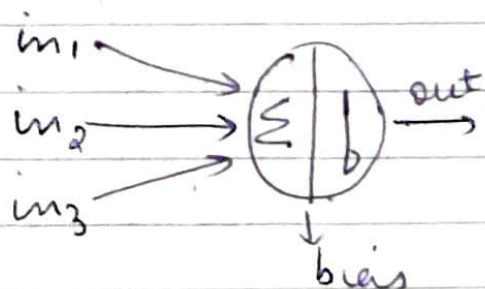
- axon connects to dendrites via synapses

it varies in strength

& it may be excitatory or inhibitory



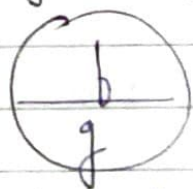
L7.7 DL Basics - 1



basic unit in
neural network is
neuron (node / unit)

- each input has a wt. which is assigned on the basis of its relative importance to other inputs.
- It has a non-linear pattern.
- perceptron

$$y \in \{0, 1\}$$



$$x_1, x_2, \dots, x_n \in \{0, 1\}$$

wt. of inputs are learnt.

$$g(x_1, \dots, x_n) = z = \sum_{i=1}^n w_i x_i$$

$$y = f(z) = 1 \text{ if } z \geq 0$$

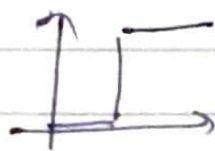
- θ is threshold & $f(\cdot)$ is non-linearity
- AND & OR func^{ns} $\rightarrow$ they are linearly separable.
- limita^{ns} of perceptron $\rightarrow$ can't work on non-linear, eg. XOR
- MLP (multi-layer perceptron) $\rightarrow$ 1 or more hidden layers.

This solves the XOR problem.

- many non-linearly separable problems were solved.

L78 DL Basis - 2

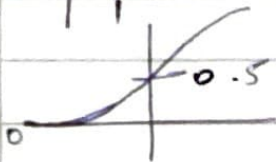
- thresholding logic of perceptron is very harsh



$$z = \sum_{i=1}^n w_i x_i$$

diff. types
of non-linearity

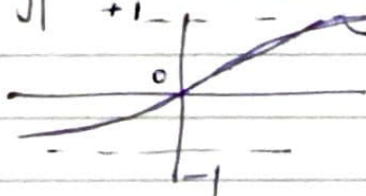
- sigmoidal funcⁿ →



$$f(z) = \frac{1}{1 + e^{-z}}$$

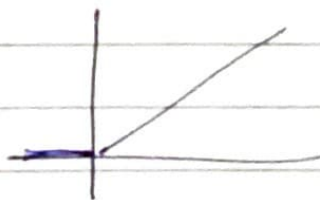
$$f(z)' = f(z)(1 - f(z))$$

- hyperbolic tangent funcⁿ → $f(z) = \tanh(z)$

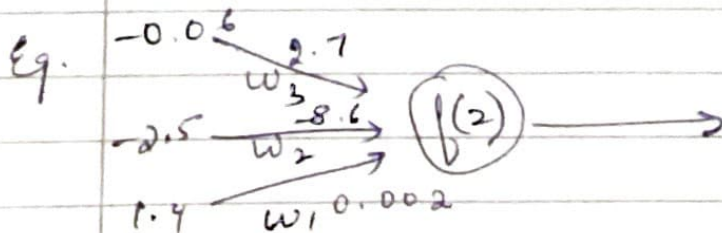


$$f(z)' = 1 - f^2(z)$$

- ReLU → $f(z) = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{if } z \geq 0 \end{cases}$



$$f(z)' = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$



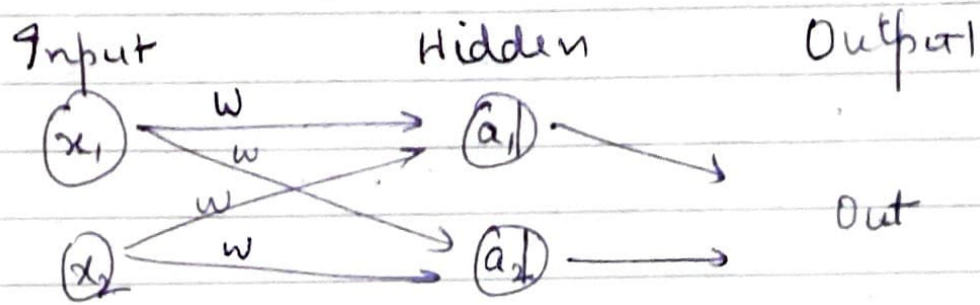
$$z = -0.06 \times 2.7 + -2.5 \times -8.6 + 1.4 \times 0.002$$

$$= 21.34$$

$$f(z) = \frac{1}{1 + e^{-z}} \approx \text{close to } 1.$$

Wts. are
impt.

- FFNN (feed forward neural network)



- multiple hidden layers, each node is a neuron.
- all wts associated with the neurons are c/d the parameters. The parameters of the network are learnt.